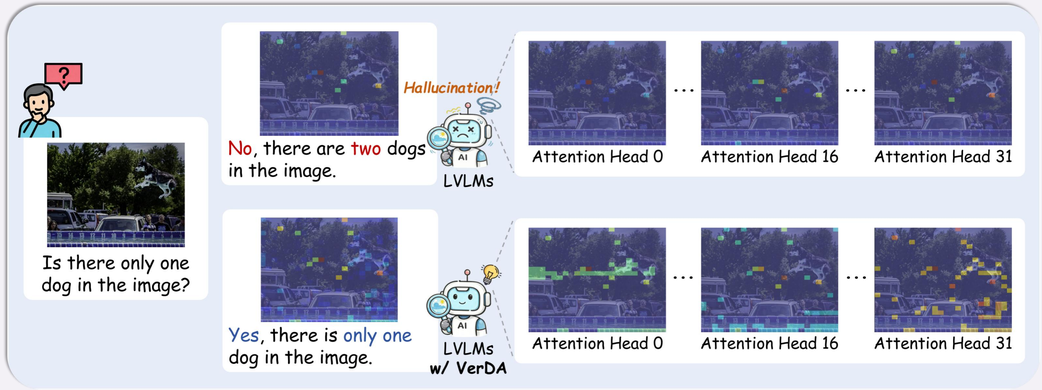
Attention Heads are Complementary Visual Units: Mitigating Hallucinations in LVLMs via Adaptive Visual Cues Focusing

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MOTIVATIONS & CONTRIBUTIONS



Limitations of Prior Work:

- **Neglected Head-Specific Visual Information.** Existing training-free methods either operate at the logits level or manipulate visual attention without fully considering *visual cues within individual attention heads.*
- **Incomplete Characterization of Visual Forgetting.** Previous studies *primarily provide a characterization* of visual forgetting as a decrease in overall visual attention during generation.
- **Insufficient Long-Range Textual Context Utilization.** The relationship between long-range *contextual information utilization and hallucination in LVLMs remains underexplored.*

Our Contributions:

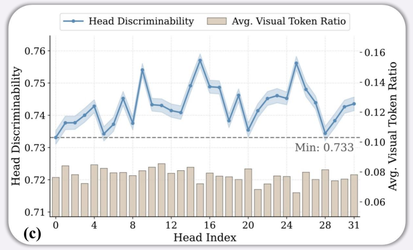
- **Insights into Hallucination Mechanisms.** We identify three findings: (i) sparse and complementary head-wise visual cues, (ii) decreasing visual cues proportion during generation, and (iii) weaker long-range textual context utilization associated with hallucination.
- **Training-Free Hallucination Mitigation Framework.** We propose VerDA, a training-free method that mitigates hallucinations.
- **Extensive Experimental Validation.** VerDA outperforms existing methods on widely-adopted hallucination benchmarks with negligible additional time cost.

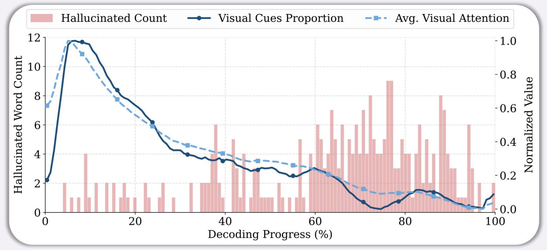
INSIGHTS

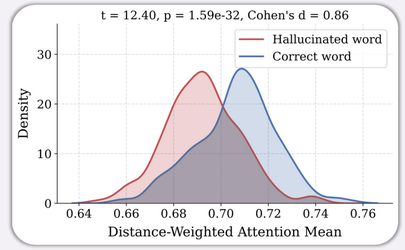
Insight 1: Visual cues are sparsely distributed within each head, while different heads capture complementary cues.

Insight 2: The proportion of visual cues within visual attention decreases during generation.

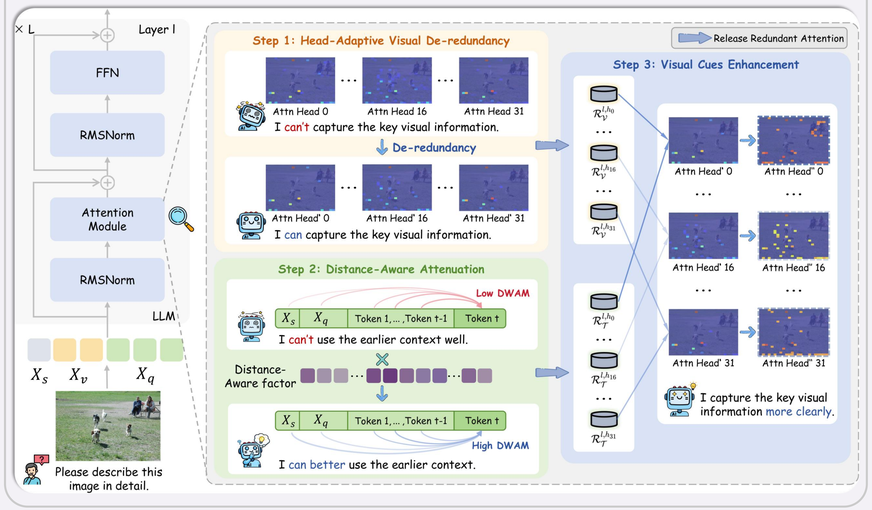
Insight 3: Hallucination in LVLMs is associated with a distance-decay effect in textual context utilization.







METHODS -- VerDA



SCAN FOR MORE

PAPER

CODE

HOMEPAGE

WECHAT

EXPERIMENTS

Method	Venue	LLaVA-1.5-7B $CHAIR_S \downarrow$	LLaVA-1.5-7B $CHAIR_I \downarrow$	LLaVA-1.5-7B $D_1 \uparrow$	LLaVA-Next-8B $CHAIR_S \downarrow$	LLaVA-Next-8B $CHAIR_I \downarrow$	LLaVA-Next-8B $D_1 \uparrow$	InstructBLIP-13B $CHAIR_S \downarrow$	InstructBLIP-13B $CHAIR_I \downarrow$	InstructBLIP-13B $D_1 \uparrow$
Beam	–	52.96$_{\pm 2.41}$	15.31$_{\pm 0.39}$	0.0624	27.94$_{\pm 1.73}$	7.57$_{\pm 0.49}$	0.0331	50.20$_{\pm 1.74}$	13.43$_{\pm 0.31}$	0.0592
CODE [19]	NeurIPS 2024	48.40$_{\pm 1.41}$	14.28$_{\pm 0.53}$	0.0692	30.42$_{\pm 1.39}$	8.03$_{\pm 0.39}$	0.0363	55.92$_{\pm 3.28}$	15.54$_{\pm 1.24}$	0.0607
VCD [16]	CVPR 2024	51.56$_{\pm 2.14}$	15.15$_{\pm 0.72}$	0.0652	32.01$_{\pm 1.57}$	8.24$_{\pm 0.91}$	0.0347	51.80$_{\pm 2.02}$	14.42$_{\pm 0.42}$	0.0602
OPERA [18]	CVPR 2024	42.96$_{\pm 1.56}$	13.20$_{\pm 0.33}$	0.0739	–	–	–	50.56$_{\pm 1.94}$	14.56$_{\pm 0.89}$	0.0639
DoLa [17]	ICLR 2024	51.56$_{\pm 2.79}$	14.79$_{\pm 1.03}$	0.0659	28.92$_{\pm 1.45}$	7.97$_{\pm 0.38}$	0.0347	51.36$_{\pm 1.79}$	14.49$_{\pm 0.78}$	0.0601
DropoutDecoding [20]	NeurIPS 2025	48.56$_{\pm 2.17}$	14.20$_{\pm 0.90}$	0.0667	28.20$_{\pm 1.39}$	7.67$_{\pm 0.42}$	0.0348	49.30$_{\pm 1.81}$	14.23$_{\pm 0.75}$	0.0542
VAF [21]	CVPR 2025	50.00$_{\pm 2.70}$	14.71$_{\pm 0.91}$	0.0668	28.96$_{\pm 1.28}$	8.08$_{\pm 0.51}$	0.0348	49.20$_{\pm 1.58}$	14.47$_{\pm 0.68}$	0.0551
SPARC [24]	ICML 2025	46.02$_{\pm 2.42}$	14.44$_{\pm 3.71}$	0.0664	31.84$_{\pm 1.76}$	8.76$_{\pm 0.74}$	0.0345	50.32$_{\pm 2.03}$	14.43$_{\pm 0.92}$	0.0602
VHR [23]	ACL 2025	33.40$_{\pm 1.56}$	9.97$_{\pm 0.51}$	0.0675	28.92$_{\pm 1.51}$	7.17$_{\pm 0.46}$	0.0361	42.48$_{\pm 1.68}$	11.98$_{\pm 0.87}$	0.0592
AdaIAT [25]	CVPR 2026	35.41$_{\pm 2.47}$	10.93$_{\pm 0.96}$	0.0813	28.04$_{\pm 1.62}$	7.97$_{\pm 1.44}$	0.0372	42.52$_{\pm 0.92}$	13.22$_{\pm 0.61}$	**0.0736**
Baseline (Greedy)	–	49.76$_{\pm 2.02}$	14.21$_{\pm 0.96}$	0.0661	29.16$_{\pm 1.27}$	7.92$_{\pm 0.68}$	0.0348	50.32$_{\pm 2.03}$	14.43$_{\pm 0.92}$	0.0602
VerDA (Ours)	–	**28.68**$_{\pm 2.09}$	**9.45**$_{\pm 0.99}$	**0.0917**	**27.52**$_{\pm 1.04}$	**6.86**$_{\pm 0.61}$	**0.0382**	**41.08**$_{\pm 2.92}$	**11.69**$_{\pm 0.83}$	0.0644

Method	LLaVA-1.5-7B Acc ↑	LLaVA-1.5-7B F1 ↑	LLaVA-Next-8B Acc ↑	LLaVA-Next-8B F1 ↑	InstructBLIP-13B Acc ↑	InstructBLIP-13B F1 ↑
Beam	84.54	85.01	89.27	89.32	81.89	83.87
CODE [19]	83.23	84.60	88.58	89.09	81.41	83.67
VCD [16]	82.76	83.92	88.31	88.67	81.99	83.91
OPERA [18]	84.67	85.47	–	–	76.27	80.35
DoLa [17]	84.03	85.10	89.09	89.46	81.22	83.59
DropoutDecoding [20]	83.97	84.99	88.99	89.38	81.34	83.56
VAF [21]	83.38	84.72	88.53	89.09	77.82	81.57
SPARC [24]	83.84	84.97	89.02	89.41	81.42	83.66
VHR [23]	84.00	85.49	89.04	89.40	82.64	84.21
AdaIAT [25]	83.85	84.99	88.91	89.36	81.45	83.69
Baseline (Greedy)	83.84	84.97	89.02	89.41	81.42	83.66
VerDA (Ours)	**84.71**	**85.55**	**89.39**	**89.71**	**83.11**	**84.63**

Model	$CHAIR_S \downarrow$	$CHAIR_I \downarrow$	POPE-Acc ↑	POPE-F1 ↑
GLM-4V-9B	39.71	8.98	89.45	89.34
w/ VerDA	**36.06** (+3.65)	**7.94** (+1.04)	**89.69** (+0.24)	**89.59** (+0.25)
Qwen2.5-VL-7B	25.03	8.46	82.95	79.94
w/ VerDA	**19.88** (+5.15)	**7.19** (+1.27)	**83.45** (+0.50)	**80.67** (+0.73)
Qwen3-VL-8B	50.68	9.97	88.71	88.62
w/ VerDA	**41.52** (+9.16)	**9.05** (+0.92)	**89.20** (+0.49)	**89.16** (+0.54)
InternVL2-8B	36.64	8.34	84.78	82.83
w/ VerDA	**29.24** (+7.40)	**7.36** (+0.98)	**84.96** (+0.18)	**82.96** (+0.13)
InternVL3-9B	30.52	7.67	89.85	89.80
w/ VerDA	**28.24** (+2.28)	**7.31** (+0.36)	**89.96** (+0.11)	**90.02** (+0.22)
InternVL3-14B	33.18	8.22	88.93	88.16
w/ VerDA	**29.84** (+3.34)	**7.78** (+0.44)	**89.63** (+0.70)	**89.09** (+0.93)

